

P09-R1 / 01    MODULY TERMOPAR I ZASILANIE

Zakupione MAX31856 XU z oferty 18805671895. Pinout ze zdjęcia: regulator i rozstaw wymagają pomiaru.

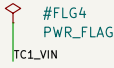
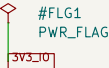
JP1/JP2 początkowo BEZ ZWOREK. 1-2: VIN=3V3 do kwalifikacji; 2-3: 5V tylko po sprawdzeniu modułu.

JP1  
VIN SELECT



J3

TC1

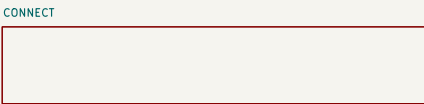
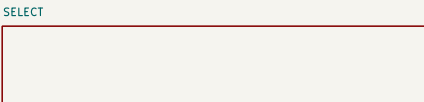
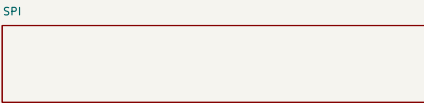
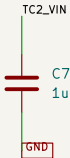
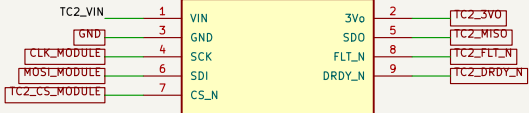


JP2  
VIN SELECT



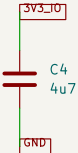
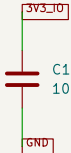
J4

TC2



3Vo obu modułów: osobne punkty pomiarowe, NIE łączyć ze sobą ani z 3V3\_I0.

Termopary K izolowane do zacisków na modułach. Nie prowadzić sygnału termopary przez nosnik P09.



Two-channel thermocouple carrier; review, hardware not tested

Prototype / schematic + PCB review

Sheet: /  
File: P09.kicad\_sch

Title: EGRLab P09 TEMP

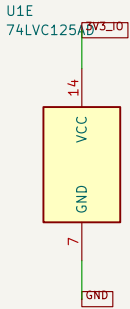
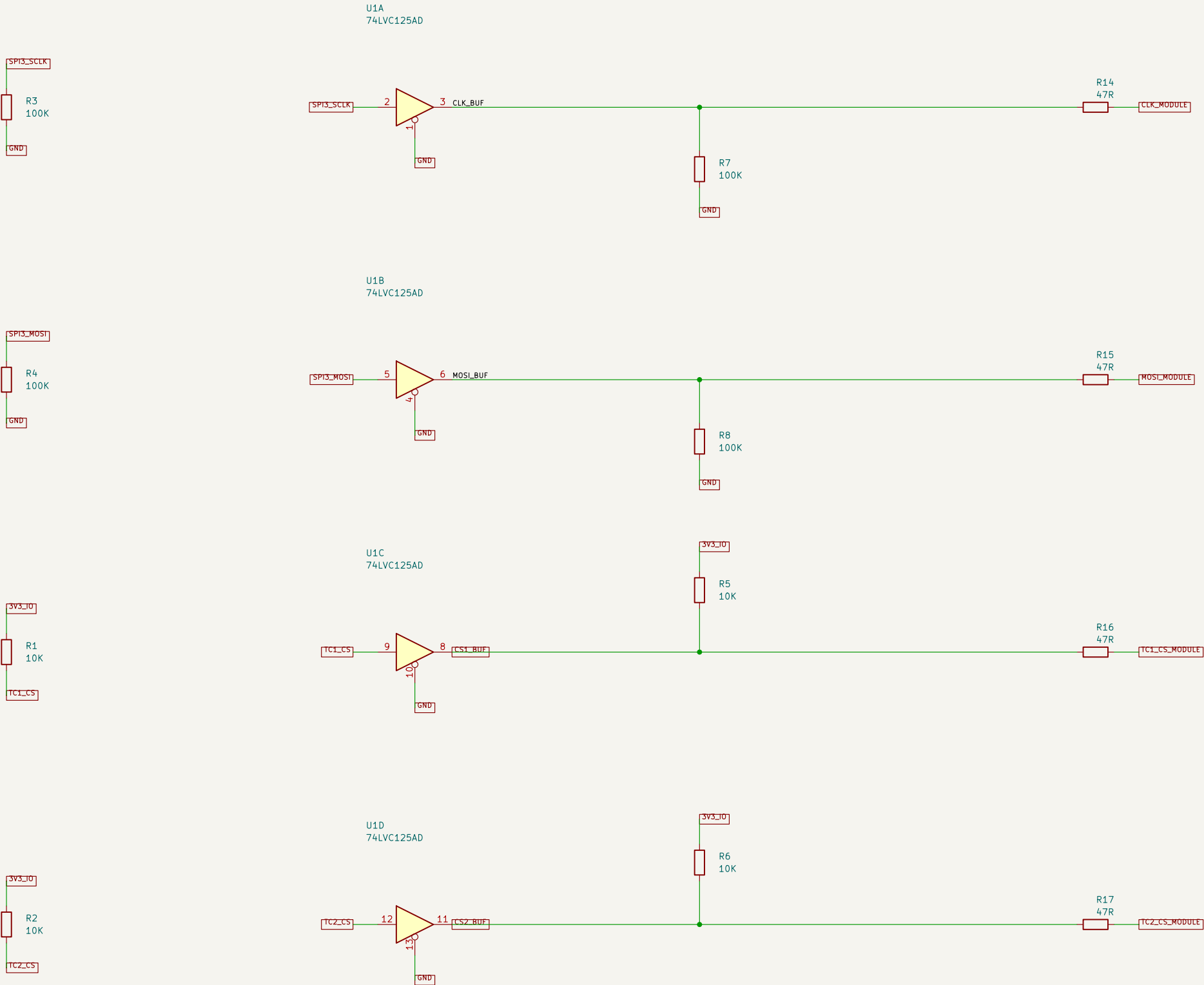
Size: A3    Date: 2026-09-27

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Rev: P09-R1

Id: 1/4

U1: Nexperia 74LVC125AD z loff, zasilanie 3V3\_I0. Nie zamieniac na HC125.



Zewnętrzne CS mają pull–up 10k; SCLK/SDI pull–down 100k. P03 U23 ma loff przy wylaczonej P03.

SPI TEMP początkowo 1 MHz, mode 1; CS setup/hold po 2 takty i przerwa obu CS HIGH przed zmianą kanału.

Two–channel thermocouple carrier; review, hardware not tested  
**Prototype / schematic + PCB review**  
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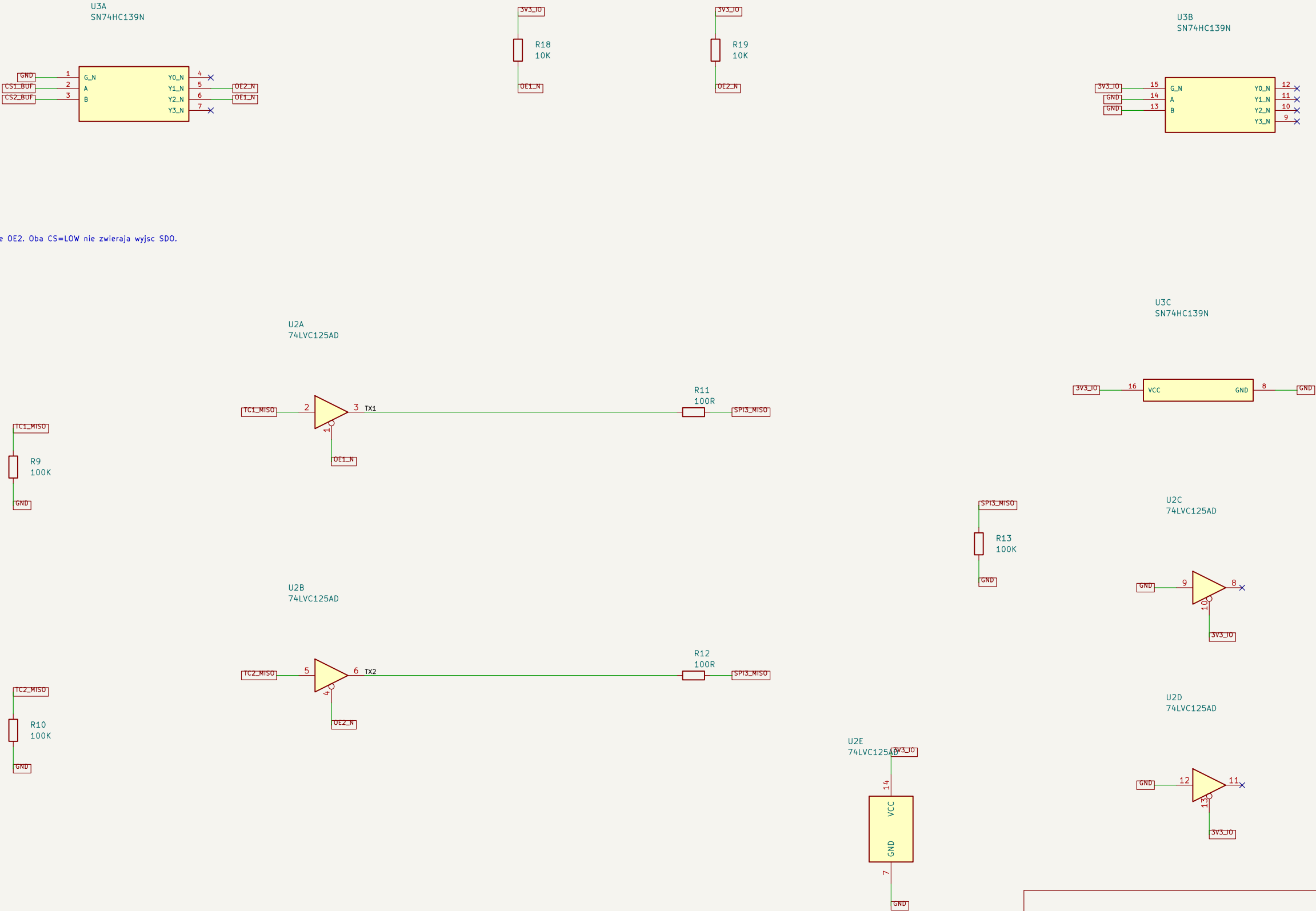
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| Size: A3                      | Date: 2026–09–27 | Rev: P09–R1 |
| KiCad E.D.A. 10.0.6           |                  | Id: 2/4     |

P09-R1 / 03    WYBOR MISO I OCHRONA WSPOLNEJ MAGISTRALI

CS1 CS2: 0 1 -> TC1; 1 0 -> TC2; 0 0 oraz 1 1 -> oba wyjscia Hi-Z.

U3: A=CS1, B=CS2; Y2 steruje OE1, Y1 steruje OE2. Oba CS=LOW nie zwieraja wyjsc SDO.

Tablica dotyczy stanów ustalonych. Przed drugim CS zapewnić oba HIGH >=1 us; 100R ograniczają prąd przejściowy.



Two-channel thermocouple carrier; review, hardware not tested

Prototype / schematic + PCB review

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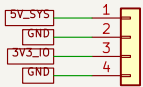
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Rev: P09-R1  
Id: 3/4

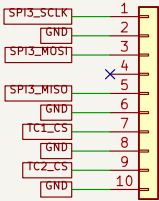
P09-R1 / 04    WIAZKI I PUNKTY POMIAROWE

LV09 -> P02-R3/J9. TEMP -> P03-R2/J7, 10p KEY4. Wiazki lutowane do PTH na P09.

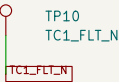
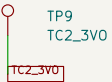
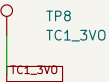
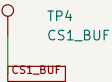
J1  
LV09



J2  
TEMP



LV09: 200 mm AWG22. TEMP: 100 mm tasma AWG28. Kotwy 12 mm od pierwszego rzędu lutu.



FLT/DRDY sa dostepne do pomiaru; dotychczasowy kontrakt TEMP nie ma dla nich wolnych zyl.

Firmware: LTCB=0x0C..0x0E; SR=0x0F. Bez poprawki stary odczyt temperatury jest bledny.

Two-channel thermocouple carrier; review, hardware not tested

Prototype / schematic + PCB review

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Title: EGRLab P09 TEMP

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| Size: A3            | Date: 2026-09-27 | Rev: P09-R1 |
| KiCad E.D.A. 10.0.6 |                  | Id: 4/4     |